



University of Colombo, Sri Lanka

University of Colombo School of Computing
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination - Semester II - UCSC AY21 [held in March/April 2025]

SCS1311 – Internet and Web Technologies

(Two (2) Hours)

Answer ALL questions

Number of Pages = 12

Number of Questions = 4

To be completed by the candidate

Index Number:

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Important Instructions to candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number **CLEARLY** on each and every page of this Question paper.
- IV. This paper consists of 4 questions in 12 pages (including the Cover Page).
- V. Answer **ALL** questions.
- VI. Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.

To be completed by the examiners

1	
2	
3	
4	
Total	

Question 01

(a) Write whether each of the given statements below is true (T) or false (F) in the given box.

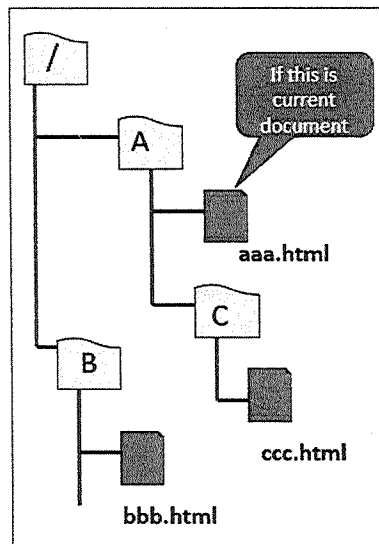
[1x20 = 20 Marks]

1.	HTML stands for HyperText Markup Language	
2.	The tag requires both src and alt attributes to be valid HTML	
3	Inline CSS styles can be added to HTML elements using the style attribute.	
4	The tag is used to insert a line break in HTML.	
5	HTML attributes can only be written in lowercase.	
6	The <iframe> tag is used to embed another HTML document within the current document.	
7	The <title> element defines the title that appears on the web page.	
8	In HTML, comments are added using <!-- comment -->.	
9	HTML elements are case-sensitive, so <DIV> and <div> are not equal tags.	
10	The tag is used to create an ordered list in HTML.	
11	A domain name is a human-readable address used to identify a website on the internet.	
12	In client-server architecture, the client requests services, and the server provides responses to those requests.	
13	Uniform Resource Locator uniquely identifies a resource on the internet	
14	A URL includes the protocol, domain name, path, and sometimes a query string or fragment identifier.	
15	The internet and the World Wide Web (WWW) are the same thing.	
16	Writing PHP code in the head section of the HTML page is not allowed.	
17	PHP is not capable of handling the business logic of a web application.	
18	Output after processing PHP content is HTML.	
19	CSS is used to validate the user input in an HTML form.	
20	JavaScript can do client-side validations.	

(b) State the difference between a dynamic website and a static website?

[1 Marks]

(c) Write down the absolute and relative paths to bbb.html and ccc.html pages assuming that you are working on aaa.html page. [4 Marks]



Absolute path for bbb.html:

Absolute path for ccc.html:

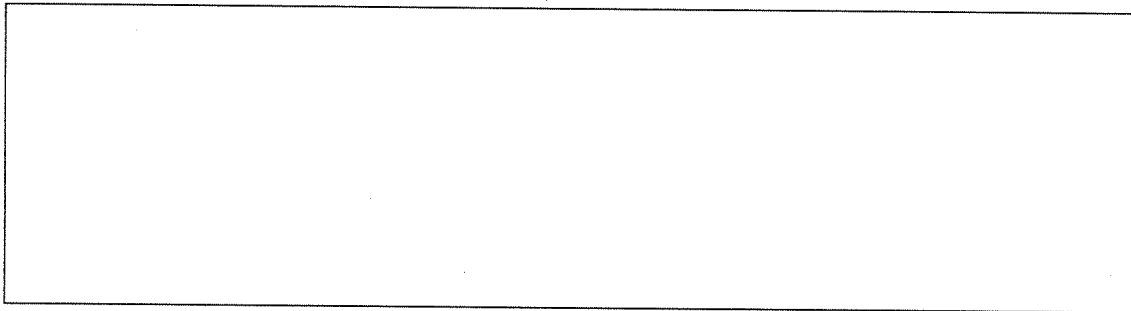
Relative path for bbb.html:

Relative path for ccc.html:

Question 02

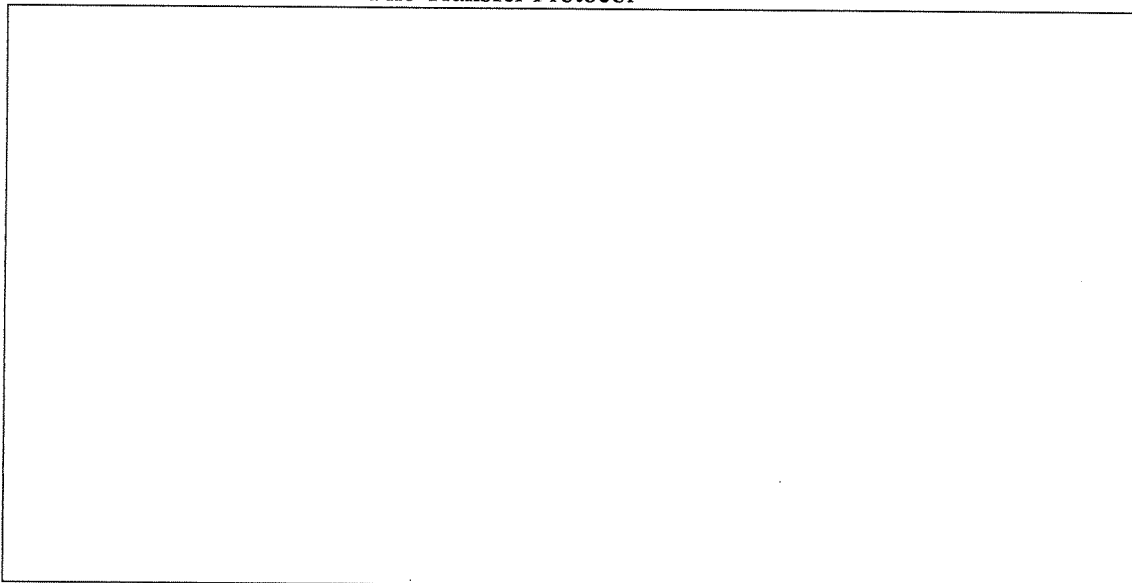
(a) Use HTML to create an image map with a circular clickable area on the image. Use the following details and write the code to implement image map. [5 marks]

- Image: "assets/images/world-map.jpg"
- Name of the map : imgMap
- Coordinates of the centre of the circular area : 250, 300
- Radius of the circular area : 10
- Link of the url to load when the circular area is clicked: <https://www.world-map.com>



(b)
i. Write the HTML code needed to create the following description list. [4 marks]

URL Uniform Resource Locator
WWW World Wide Web
FTP File Transfer Protocol



ii. Draw the output of the following HTML code.

[8 marks]

```
<html>
  <TABLE border="1">
    <TR>
      <TD colspan="2" rowspan="2">Cell 1</TD>
      <TD>Cell 2</TD>
    </TR>
    <TR>
      <TD>Cell 3</TD>
    </TR>
    <TR>
      <TD>Cell 4</TD>
      <TD>Cell 5</TD>
      <TD>Cell 6</TD>
    </TR>
    <TR>
      <TD>Cell 7</TD>
      <TD colspan="2">Cell 8</TD>
    </TR>
  </TABLE>
</html>
```


Question 03

(a) List down **four** (4) main Object-Oriented Programming (OOP) concepts.

[4 marks]

(b) Explain the difference between the class and the object. Provide an example.

[4 marks]

(c) What is a JavaScript “event”? Give **three** (4) examples for valid JavaScript events associated with HTML buttons.

[6 marks]

(d) Answer the given questions based on the JavaScript code given below.

```
function addPrice(amount) {  
    with(this) {  
        price = amount;  
    }  
}  
function book(title, author) {  
    this.title= title;  
    this.author= author;  
    this.price= 0;  
    this.addPrice= addPrice;  
}
```

- i. Write the JavaScript code to create a new book titled "Madolduwa" by Martin Wickramasinghe. (Note: use the variable named myBook to store the newly created book object). [5 marks]

- ii. Write the JavaScript code to add the book's price as 2500 by calling the relevant function. [3 marks]

- iii. Write the JavaScript code to print the book details to generate the following output. [3 marks]

```
Book title is : Madolduwa  
Book author is : Martin Wickramasinghe  
Book price is : 100
```

Question 04

(a) Write the outputs of the following php code snippets in the given space.

[2x5 = 10 marks]

i)	<pre><?php \$numbers = array(10, 20, 30, 40, 50); \$total = 0; foreach (\$numbers as \$number) { if (\$number > 25) { \$total += \$number; } } echo "Total of numbers greater than 25: \$total"; ?></pre>	
ii)	<pre><?php \$models = array("Toyota" => array("Fortuner", "Camry"), "Nissan" => array("Leaf", "Sylphy"), "Hyundai " => array("Sonata ", "Tucson")); foreach (\$models as \$make => \$names) { echo "\$make Models
"; foreach (\$names as \$name) { if(\$name == "Leaf"){ echo "- \$name (electric)
"; } echo "- \$name
"; } } ?></pre>	
iii)	<pre><?php function add(\$n1=20, \$n2=40){ \$n3=\$n1+\$n2; echo "Addition is: \$n3
"; } add(); add(10); add(40,40); ?></pre>	

iv)	<pre> <?php \$i = 1; while (\$i <= 5) { \$j = 1; while (\$j <= \$i) { echo \$j . " "; \$j++; } echo "
"; \$i++; } ?> </pre>	
v)	<pre> <?php \$average = 85; switch (\$average) { case (\$average >= 90): echo "Excellent
"; case (\$average >= 80): echo "Good
"; case (\$average >= 70): echo "Satisfactory
"; default: echo "Needs Improvement"; } ?> </pre>	

- (b) Following HTML code segment is written for creating a new user in a web application. The form simply accepts user name and password to register a new user.

```
<html>
  <head>
    <title>Create User</title>
  </head>
  <body>
    <h3>Create New User</h3>

    <form method="post" action="processor.php" name="registration">
      <input type="text" name="name" placeholder="Name">
      <input type="text" name="password" placeholder="Password">

      <input type="submit" name="create" value="Create User">
    </form>
  </body>
</html>
```

In order to validate whether the fields are empty using JavaScript, you need to access the values entered by the user. Write down a JavaScript function to temporarily hold the values of the name and password fields and validate for empty values.

[4 marks]

- (c) The form submits user data to “processor.php” file as mentioned in the “action” attribute of the form. Write down the php code to save form submitted data inside processor.php file. Note: save form values to \$name and \$password variables in the processor.php file.

[4 marks]

(d) “processor.php” file needs to connect to the database in order to save data to the database table. Create a database connection using the following connection details.

- Host: localhost
- User name: user
- Password: user@123
- Database name: empDatabase

[4 marks]

(e) Write the MySQL query string to save data to the “users” table in the database. Assign the sql statement to *\$sql* variable. Note: the users table has *uname* and *pwd* fields.

[2 marks]

(f) Use the connection you created in 4 (c) to save data to the users table using the query written in 4 (d).

[1 marks]
